

V4.2 Toy Model 3D

Adding Potentials as Position-Dependent Phase Rotation Rate

Rev A - Exploratory Mathematical Compatibility Report

Date: 2026.05.25 | Continuation after Toy Model 3C

Status: Toy Model 3D tests whether a potential-energy field can be represented as a position-dependent shift in hidden phase-rotation rate. It does not derive the physical origin of force fields.

1. Purpose

Toy Model 3C recovered the structural form of the free-particle kinetic term from spatial phase twisting. Toy Model 3D adds the next standard quantum ingredient: a potential-energy term $V(x)$.

The narrow target is to show that a real position-dependent potential acts as a local phase-rotation-rate shift and appears as an additive Hermitian multiplication operator in the Schrodinger equation.

2. Starting point from 3C

The free-particle structure was:

$$i \hbar \frac{d\psi}{dt} = -(\hbar^2 / 2m) \frac{d^2\psi}{dx^2}$$

The kinetic operator was:

$$\hat{T} = -(\hbar^2 / 2m) \frac{d^2}{dx^2}$$

Now introduce a real potential-energy function $V(x)$.

3. Potential as local phase-rotation rate

From 3B, energy corresponds to phase-rotation rate:

$$E = \hbar \omega$$

Therefore a local potential energy $V(x)$ corresponds to a local additional rotation rate:

$$\omega_V(x) = V(x) / \hbar$$

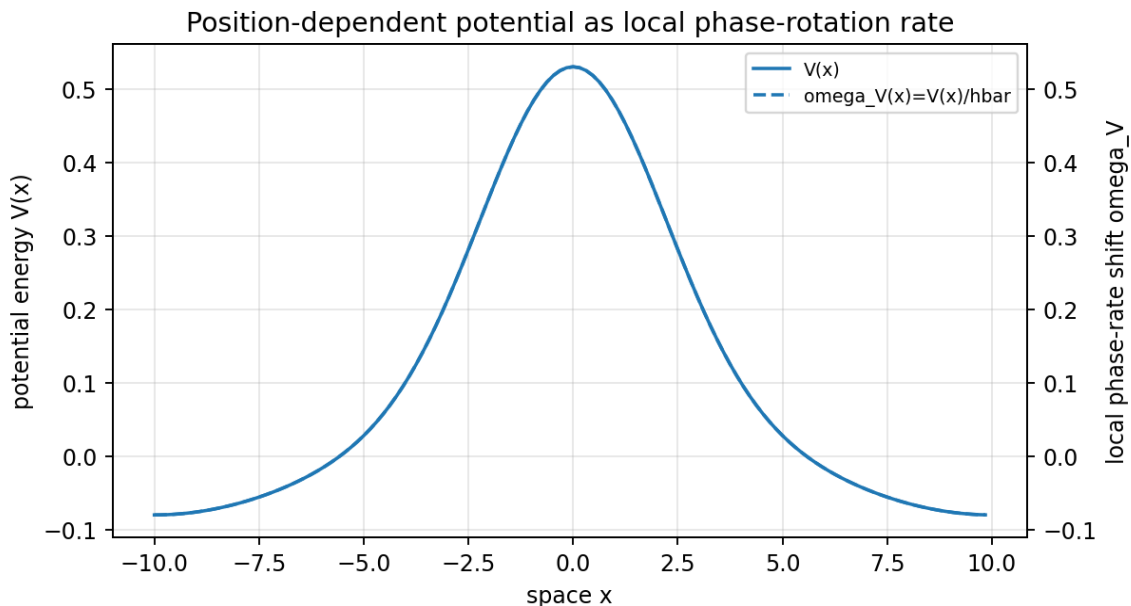
For potential-only evolution:

$$\frac{d\psi}{dt} = -i [V(x)/\hbar] \psi$$

Multiplying by $i \hbar$ gives:

$$i \hbar \frac{d\psi}{dt} = V(x) \psi$$

Plain meaning: where $V(x)$ is higher, the hidden transverse phase rotates faster; where $V(x)$ is lower, it rotates slower.



4. Full kinetic-plus-potential structure

Adding the kinetic structure from 3C and the local phase-rate shift gives:

$$i \hbar \frac{d\psi}{dt} = \left[-(\hbar^2 / 2m) \frac{d^2}{dx^2} + V(x) \right] \psi$$

This is the standard one-dimensional nonrelativistic Schrodinger structure with a scalar potential.

Term reminders:

- $V(x)$ is potential energy at position x .
- Scalar potential means it multiplies the wavefunction without changing internal components.
- Hermitian means the operator preserves probability under closed time evolution.
- The Hamiltonian H is the total energy operator: kinetic plus potential.

$$H = T + V$$

$$T = -(\hbar^2 / 2m) \frac{d^2}{dx^2}$$

$$V = \text{multiplication by real } V(x)$$

5. Hermiticity and norm preservation

If $V(x)$ is real, then the potential operator is Hermitian. The kinetic operator is also Hermitian under standard boundary conditions. Therefore:

$$H = T + V \text{ is Hermitian}$$

The time-evolution generator is:

$$A = -i H / \hbar$$

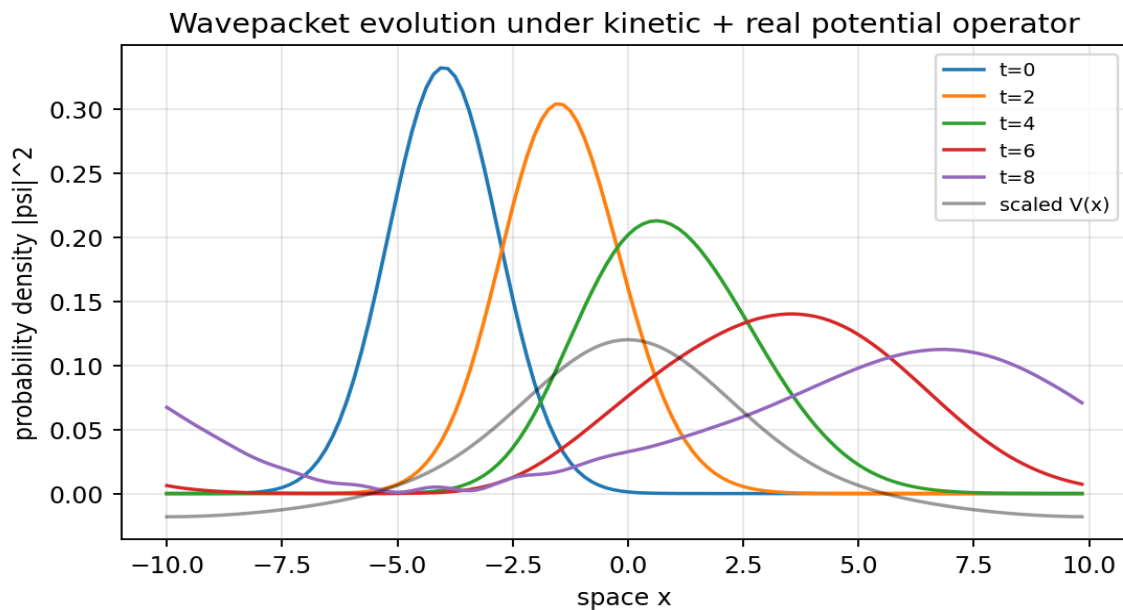
If H is Hermitian, then A is anti-Hermitian. Anti-Hermitian generators preserve norm:

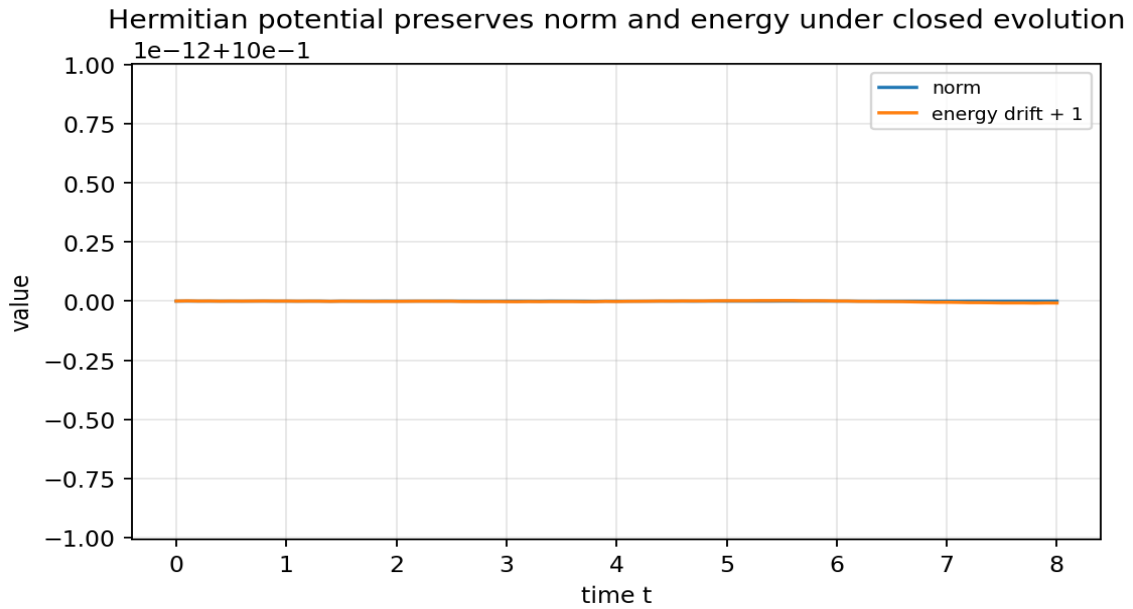
$$d/dt \langle \psi | \psi \rangle = 0$$

This is the formal probability-conservation check. In 6D language, a real potential shifts local hidden phase rotation but does not create or destroy probability.

6. Numerical wavepacket test

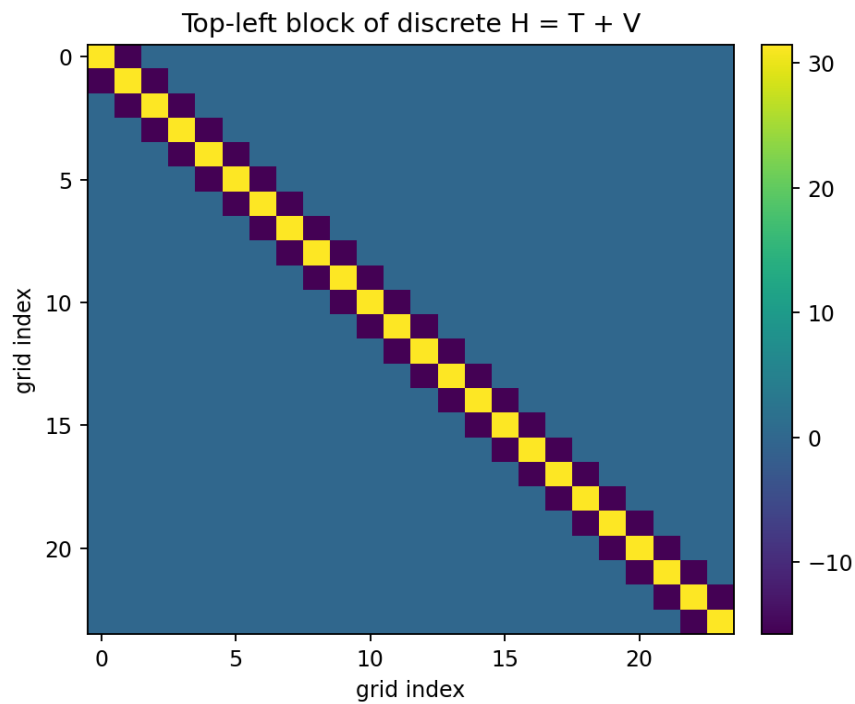
A periodic 1D grid was used with a real potential. The Hamiltonian $H = T + V$ was diagonalized and used for exact closed-system unitary evolution.





7. Discrete operator structure

The numerical Hamiltonian is a real symmetric matrix. The off-diagonal bands come from the second-derivative kinetic term. The diagonal contribution includes $V(x)$.



8. Numerical checks

Check	Result
Grid points	128
Potential Hermiticity error	0.000e+00
Kinetic Hermiticity error	0.000e+00
Hamiltonian Hermiticity error	0.000e+00
Generator anti-Hermiticity error	0.000e+00
Max norm drift under evolution	4.441e-16
Max energy drift under evolution	8.438e-15
Max norm-derivative error	3.331e-15
Max imaginary energy expectation	1.665e-15

Check	Result
Constant-potential plane-wave residual	3.662e-15

9. Interpretation inside the 6D Projection Framework

Toy Model 3D supports the following limited interpretation:

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position-dependent relational influence
-> local hidden transverse phase-rate shift
->  $\omega_V(x) = V(x)/\hbar$ 
-> additive term  $V(x) \psi$ 
->  $H = \text{kinetic phase-twist term} + \text{local phase-rate term}$ 

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This converts the potential term into a geometric phase-rate modulation in the hidden transverse temporal plane.

10. What Toy Model 3D achieves

- It represents $V(x)$ as a position-dependent hidden phase-rotation-rate shift.
- It recovers the additive scalar potential term $V(x) \psi$.
- It verifies that real potentials are Hermitian multiplication operators.
- It verifies that $H = T + V$ preserves probability under closed-system evolution.
- It extends the free-particle structure of 3C to the standard kinetic-plus-potential Schrodinger form.

11. What Toy Model 3D does not achieve

- It does not derive the physical origin of $V(x)$.
- It does not derive electromagnetic, gravitational, nuclear, or other force fields.
- It does not derive mass m or Planck's constant $\hbar$.
- It does not derive relativistic quantum mechanics.
- It does not derive quantum field theory or the Standard Model.
- It does not prove physical 3T existence.

12. Close-out statement for 3D

Toy Model 3D successfully adds scalar potentials to the reciprocal-projection phase-dynamics track. A real potential $V(x)$ is structurally equivalent to a local hidden transverse phase-rotation-rate shift $\omega_V(x)=V(x)/\hbar$. This produces the additive Schrodinger term $V(x)\psi$. Combined with the kinetic structure from 3C, it gives $i \hbar \text{d}\psi/\text{d}t = [-(\hbar^2/2m)\text{d}^2/\text{d}x^2 + V(x)]\psi$. Hermiticity and norm preservation are maintained when $V(x)$ is real. This is a structural compatibility result, not a derivation of actual force fields.

13. Recommended next chapter

The next logical chapter is Toy Model 3E: Gauge Phase and Electromagnetic Coupling as Local Phase-Connection Structure.

1. Introduce local phase transformations $\psi \rightarrow \exp(i \chi(x,t)) \psi$.
2. Show why ordinary derivatives must become covariant derivatives when local phase can vary.
3. Recover the structural form of electromagnetic gauge potentials A and ϕ .
4. Check norm preservation and gauge-invariant probability density.
5. Keep guardrails: do not claim derivation of electromagnetism until physical source equations are addressed.

14. Rebind prompt for continuation

We are resuming after Toy Model 3D.

3D tested scalar potentials as position-dependent hidden phase-rotation-rate shifts.

From 3B:
 $E = \hbar \omega$

Potential contribution:
 $\omega_V(x) = V(x) / \hbar$

Potential-only evolution:
 $\text{d}\psi/\text{d}t = -i [V(x)/\hbar] \psi$

Therefore:

$$i \hbar \frac{d\psi}{dt} = V(x) \psi$$

Combined with 3C kinetic structure:

$$i \hbar \frac{d\psi}{dt} = [-(\hbar^2/2m)d^2/dx^2 + V(x)] \psi$$

Hermiticity:

If $V(x)$ is real, the potential operator is Hermitian.

$H = T + V$ is Hermitian.

$A = -iH/\hbar$ is anti-Hermitian.

Norm is preserved.

Guardrail:

3D does not derive actual force fields, mass, $\hbar$, relativistic QM, QFT, the Standard Model, or physical 3T existence.

Next recommended chapter:

Toy Model 3E - Gauge Phase and Electromagnetic Coupling as Local Phase-Connection Structure.